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What is a Good School, and Can Parents Tell? Evidence on the Multidimensionality of School Output

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1 Big picture

- **Central question.** The paper asks whether a school that causally raises high-stakes test scores is also a school that improves other outcomes that families and policymakers care about, and whether parents can identify and choose schools with those causal impacts.
- **Core empirical setting.** The setting is Trinidad and Tobago’s centralized secondary-school assignment system. Students take the Secondary Entrance Assessment (SEA), submit ranked school lists, and are assigned mostly through a Deferred Acceptance mechanism using choices and SEA scores.
- **Key data advantage.** The authors link students’ school choices and administrative school assignments to many later outcomes: low-stakes exams, high-stakes school-leaving exams, tertiary-exam outcomes, dropout, police arrests, birth records, and formal labour-market participation.
- **Identification idea.** Conditional on choice-group fixed effects and smooth functions of SEA scores, the assignment rule creates quasi-random variation in which school a student attends. The assigned school is used as an instrument for actual attendance.
- **First main result.** Individual schools have meaningful causal impacts on many dimensions of output. One standard deviation in persistent school effects is large for high-stakes tests, low-stakes tests, dropout, teen motherhood, arrests, and formal employment.
- **Second main result.** School impacts are multidimensional. High-stakes test value added is only weakly related to school impacts on several non-test outcomes. A high-test-score school is not necessarily the same as a school that reduces arrests, reduces teen motherhood, or increases formal employment.
- **Third main result.** Parents choose schools that improve high-stakes test scores, even after conditioning on peer quality and average outcomes. This contrasts with evidence from some larger, lower-information settings.
- **Fourth main result.** Parents also choose schools that reduce teen arrests and teen motherhood and increase formal labour-market participation. Low-achieving students’ parents appear relatively more responsive to non-academic school impacts, while high-achieving students’ parents appear relatively more responsive to academic impacts.
- **Policy implication.** Test-score-only accountability can misclassify school quality. If schools produce multiple outputs, policies based only on test scores can create multitasking distortions and may not improve the outcomes parents value.

2 Incremental contribution of the paper

2.1 Contribution relative to school value-added research

- Prior school value-added work typically focuses on test scores or on a limited set of schools, such as oversubscribed charters or exam schools.
- This paper estimates causal impacts for 132 individual public secondary schools, covering 98.5% of the public secondary schools in Trinidad and Tobago.

- The paper estimates causal school impacts not only on academic outcomes, but also on criminality, teen motherhood, dropout, and formal labour-market participation.

Interpretation. The first incremental contribution is scale plus breadth. The paper moves from measuring a small number of school effects on test scores to measuring a representative set of individual-school causal effects on multiple student outcomes.

2.2 Contribution relative to school choice and parent-preference research

- Existing work often asks whether parents value school average achievement or test-score value added.
- This paper asks whether parents value schools' *causal impacts* on a broad menu of outcomes.
- The paper explicitly links estimated school causal effects to parents' ranked school choices.
- It does so in a relatively high-information setting with fewer schools than very large urban districts, reducing the concern that parents are overwhelmed by too many options.

Interpretation. The second incremental contribution is preference measurement. The paper does not infer that parents value a school merely because the school has high average scores; it asks whether parents choose schools that causally improve outcomes after controlling for peer quality, averages, and distance.

2.3 Contribution relative to discrete-choice modelling under Deferred Acceptance

- Standard rank-ordered logit models assume that submitted rankings reveal true preference order.
- Under Deferred Acceptance with a limited list of choices, students may rank strategically because admission probabilities differ across schools.
- The paper modifies the exploded multinomial logit framework to allow choices to reflect both ex post school utility and admission probabilities.
- The approach avoids treating every omitted school as necessarily worse than every listed school.

Interpretation. The third contribution is methodological: a choice model tailored to limited-choice school assignment systems, where strategic application behaviour is plausible.

2.4 Contribution to policy debates

- Test-score accountability assumes that test score value added is a sufficient measure of school quality.
- School-choice policy assumes that parents can identify and choose good schools.
- The paper shows that both assumptions are incomplete: school quality is multidimensional, but parents may detect some non-test dimensions of school quality.

Takeaway. The paper’s incremental contribution is not a single new coefficient. It is the combination of representative causal school effects, multidimensional outcomes, parent preferences, and a choice model suited to strategic rank lists.

3 Data and institutional setting

3.1 Trinidad and Tobago secondary-school assignment

- Students sit the SEA at the end of primary school, around age 11.
- Families submit ranked lists of preferred secondary schools.
- Most students are assigned by a centralized algorithm based on SEA scores, submitted rankings, and school capacities.
- The relevant margin for identification is that students with similar choices and similar SEA scores can be assigned to different schools because of school-specific admission cutoffs and choice-group variation.

Interpretation. The system creates a policy-generated source of school attendance variation. Because the assignment algorithm uses known rules, the econometrician can condition on the assignment inputs and exploit residual quasi-random variation.

3.2 Administrative datasets

- **SEA records:** application records, student identifiers, SEA scores, and ranked lists of secondary-school choices for cohorts 1995–2012.
- **NCSE records:** low-stakes national exams taken roughly three years after secondary-school entry, available for cohorts 2006–2012.
- **CSEC records:** high-stakes school-leaving exams taken roughly five years after entry, available for cohorts 1995–2011.
- **CAPE records:** tertiary-level certification exam outcomes, available for cohorts 1999–2009.
- **Criminal records:** official police arrest records, linked for cohorts 1995–2010.
- **Birth records:** national birth registry records, used to identify teen motherhood, linked for cohorts 2004–2010.
- **Labour-market records:** social-security contribution records, used to measure formal employment, linked for cohorts 1995–2006.

Interpretation. The paper’s data allow it to follow students well beyond immediate school performance. This is essential because the main claim is that school quality cannot be summarized by test scores alone.

3.3 Main outcome variables

- **Academic outcomes:** standardized high-stakes index, standardized low-stakes index, dropout by age 14, CSEC participation and performance, CAPE participation and performance.
- **Non-academic outcomes:** no live birth by age 19, not arrested by age 18, formal employment at age 27 or later.
- **School-choice outcomes:** rank-ordered lists of secondary schools submitted by families.
- **School attributes in choice models:** school causal impacts, school average outcomes, peer quality, distance or proximity, gender composition, island indicators, and admission probabilities.

Interpretation. The outcomes are deliberately multidimensional. The paper treats schools as producers of cognitive skills, educational attainment, social behaviour, fertility outcomes, and labour-market attachment.

4 Models and empirical specifications

4.1 Conceptual school-effect model

The objective is to estimate the causal effect of attending school j relative to other relevant schools. A stylized potential-outcome representation is

$$Y_i(j) = \theta_j + m_i(j) + \eta_i, \quad (1)$$

where θ_j is fixed school value added, $m_i(j)$ is a match effect between student i and school j , and η_i is student-specific unobserved potential performance.

Interpretation. The paper’s main estimates are interpretable as school causal effects if assignment is conditionally exogenous and if average match effects do not vary in ways that break additivity across school comparisons.

4.2 Assignment variation

The paper uses two sources of variation generated by the assignment mechanism.

1. **Discontinuity variation:** students just above and below a school’s cutoff have similar SEA scores and choices but different school assignments.
2. **Across-cutoff or difference-in-differences variation:** because the same comparison school can appear in multiple choice groups, the model can compare changes across cutoffs after controlling for choice groups and smooth SEA functions.

Interpretation. Cutoff variation gives a clean local comparison. Across-cutoff variation helps identify school effects more broadly across the distribution, which is important because the paper wants school-level value-added estimates for nearly all public secondary schools.

4.3 Reduced-form school assignment model

The paper estimates school-assignment and outcome models that condition flexibly on SEA scores and choice groups. The core two-stage idea is captured by the following equations:

$$I_{ij} = \sum_{\tau} (I_{i\tau} \pi_{\tau j}) + f_j(SEA_i) + \lambda_{jc} + X_i' \delta_j + S_{jt} + u_{ijct}, \quad (2)$$

$$Y_{ijct} = \sum_j \hat{I}_{ij} \theta_j^{TOTIV} + f(SEA_i) + \lambda_c + X_i' \delta + S_t + \varepsilon_{ijct}. \quad (3)$$

secondary schools in the nation (i.e. 132 schools). The resulting two-stage least squares (2SLS) model is as follows:

$$I_{i,j} = \Sigma(I_{i,\tau} \cdot \pi_{\tau j}) + f_j(SEA_i) + \lambda_{j,c} + \mathbf{X}_{it}' \delta_j + S_{j,t} + v_{ijct}, \quad \text{for each } j \in J \quad (5)$$

$$Y_{ijct} = \Sigma(\hat{I}_{i,j} \cdot \theta_j^{TOTIV}) + f(SEA_i) + \lambda_c + \mathbf{X}_{it}' \delta + S_t + \epsilon_{ijct} \quad (6)$$

Original paper excerpt: 2SLS school-impact equations.

Interpretation. The first equation predicts attendance at school j using algorithm-based school assignment. The second equation uses predicted attendance to estimate each school's causal impact on an outcome. Choice-group fixed effects compare students with the same ranked lists, and smooth SEA functions absorb achievement gradients.

4.4 Persistent school-effect variance

Because school effects are estimated with sampling error, the paper does not simply take the raw variance of estimated effects. Instead, it estimates school effects separately for even and odd SEA cohorts:

$$\hat{\theta}_{j,p}^{TOTIV} = \theta_j^{TOT} + \mu_{jp}, \quad p \in \{even, odd\}. \quad (4)$$

The covariance between even- and odd-cohort estimates identifies the variance of the persistent component:

$$Var(\theta_j^{TOT}) \approx Cov(\hat{\theta}_{j,even}^{TOTIV}, \hat{\theta}_{j,odd}^{TOTIV}). \quad (5)$$

Interpretation. This separates persistent school quality from transitory cohort noise. Without this correction, the dispersion of estimated school effects would exaggerate true school-quality variation.

4.5 Preference model under limited Deferred Acceptance choices

Parents submit a limited number of ranked choices. The paper distinguishes the ex post utility of attending a school from the ex ante utility of listing a school when admission probabilities differ.

The ex post utility of attending school j is

$$U_{ij} = \delta(X_i, Z_{ij}) + \varepsilon_{ij}. \quad (6)$$

In the empirical choice model, this becomes

$$U_{ij} = \beta' Z_{ij} + \varepsilon_{ij}. \quad (7)$$

To allow for strategic behaviour, the paper lets admission probabilities enter the utility of ranking a school:

$$U_{ij|p_{ij}} = \beta'_1 Z_{ij} + \beta'_2 (Z_{ij} q_{ij}) + \pi q_{ij} + \varepsilon_{ij}, \quad (8)$$

where p_{ij} is the probability of admission and $q_{ij} = 1 - p_{ij}$ is the rejection probability.

Condition (2): $U_{ij}^{r_1} | p_{ij} > U_{ik}^{r_m} | p_{ik} \forall k \neq j \in J$: Conditional on the admission probabilities, parent i *ex post* prefers their first-ranked school over any school not in the submitted choices.

The two conditions suggest that, where R_i is the maximum number of alternatives ranked by parent i , assuming rational choices, the probability that a parent i submits a particular ranking over all schools is given by equation (10) below.

$$Prob[r_{i1}, r_{i2}, \dots, r_i] = Pr \left[\begin{array}{l} \underbrace{(U_{ij}^{r_1} | p_{ij} > U_{ik}^{r_m} | p_{ik} \forall k \neq j \in J)}_{\text{Top choice preferred to all non-chosen schools conditional on admission probabilities}} \\ \cap \underbrace{(U_{ij}^{r_1} > U_{ik}^{r_m}, 1 < m, \forall m \in \{2, \dots, R_i\}) \cap \dots \cap (U_{ij}^{r_{R_i-1}} > U_{ik}^{r_{R_i}})}_{\text{Higher ranked chosen ex-post schools are preferred to lower-ranked chosen schools}} \end{array} \right]. \quad (10)$$

Therefore, if one had measures of the admission probabilities, and one correctly modelled how admission probabilities influence choices, then one can infer *ex post* preferences across all schools based on the choices (even when the top choice is not the *ex post* preferred school).

5.2. Modified exploded multinomial logistic model

Equation (10) defines the likelihood of observing a set of choices as a function of parent utilities for schools and random errors. We make some assumptions on the form of U_{ij} , the form of $U_{ij|p_{ij}}$, and the distribution of ε_{ij} to model equation (10) and use the observed choices to infer parental preferences for school attributes. Following [Hastings et al. \(2009\)](#) and [Abdulkadiroğlu et al. \(2020\)](#), we parametrize $\delta(\cdot)$ as a linear-in-parameters function of the school characteristics. Where β is a vector of deterministic components of school preferences, the *ex post* utility of student i from attending School j is

$$U_{ij} = \beta' Z_{ij} + \varepsilon_{ij} \quad (11)$$

To model strategic behaviours, we also parametrize $U_{ij|p_{ij}}$. In a simple behavioural model, when choosing the top choice school, parents may trade-off admission probability against other school attributes. This probability may enter the *ex ante* utility additively. Alternatively, as suggested in [Chade and Smith \(2006\)](#), individuals may make choices based on expected utility such that the admission probability is multiplicative to the *ex post* utility. To allow for *both* possibilities, we implement a flexible model that includes the admission probabilities as a standalone predictor of choices and also includes interactions of each observed characteristics with these probabilities. In many research settings, this probability is difficult to uncover. Fortunately, because we have many years of admissions data and students are assigned to schools based on a known algorithm, we can approximate this probability with the historical likelihood that student i would have scored above the cutoff for school j given their own incoming SEA score. We discuss these estimated probabilities in [Supplementary Appendix I](#). For ease of exposition, we model behaviours with respect to $q_{ij} = 1 - p_{ij}$ —i.e., the probability of rejection of person i from School j . It follows that

$$U_{ij|p_{ij}} = \beta'_1 Z_{ij} + \beta'_2 (Z_{ij} q_{ij}) + \pi q_{ij} + \varepsilon_{ij}. \quad (12)$$

Original paper excerpt: choice probability and modified exploded multinomial logit.

Interpretation. The model allows a parent to rank a school not only because it is intrinsically preferred, but also because admission is feasible. This matters because low-scoring students may strategically list less selective schools even if they would prefer more selective schools *ex post*.

5 Identification and credibility

5.1 Conditional exogeneity of assignment

The key assumption is that, conditional on the information used by the assignment mechanism, algorithmic assignment is unrelated to potential outcomes. The paper conditions on:

- smooth functions of incoming SEA scores;
- choice-group fixed effects, which group students with the same ranked school lists;
- individual characteristics measured at SEA registration;
- year/cohort effects and additional controls.

Interpretation. Conditional on these variables, assignment to a school is generated by the centralized rule rather than by unobserved parental or student traits.

5.2 Why choice-group fixed effects matter

- Students who list the same schools in the same order likely have similar latent preferences and constraints.
- Choice-group fixed effects compare students within the same revealed choice set.
- This reduces bias from comparing students whose families considered very different schools.

Interpretation. Choice groups are a crucial design element. They make the comparison closer to students competing for the same seats rather than comparing all students nationally.

5.3 Additivity of school effects

To compare all schools on a common scale, relative school effects must be additive. If school m relative to k equals school m relative to l plus school l relative to k , then pairwise estimates can be combined into a national school-value-added ranking.

Interpretation. Additivity is required because the assignment system generates many local comparisons rather than one universal randomized lottery. The paper reports empirical checks supporting this condition.

5.4 Noncompliance and IV interpretation

Some students do not attend their assigned school. The paper therefore uses assignment as an instrument for actual attendance.

- The first stage comes from algorithmic assignment to schools.
- The second stage estimates the effect of actual school attendance.
- The resulting estimates are treatment-on-the-treated IV school effects.

Interpretation. The IV strategy handles noncompliance so long as assignment affects outcomes through attendance and the assignment instrument is conditionally as-good-as-random.

5.5 Remaining limitations

- The school effects are relative effects; each school is compared to a set of counterfactual schools generated by the assignment mechanism.
- Preference estimates reveal the attributes of schools parents choose, not directly stated preferences.
- Parents may choose based on information or reputation correlated with causal impacts, rather than observing causal impacts directly.
- Formal employment is measured through social-security contribution records, so informal work is not fully observed.

6 Tables and figures

6.1 Table 1 and Figure 1: data and identifying variation

Read. Table 1.

- The SEA sample includes 329,481 individuals from cohorts 1995–2012.
- The population is roughly balanced by gender: 50.46% female.
- In the CSEC data, 73.22% take at least one subject, 34.55% qualify for tertiary education, and 313,580 individuals are linked.
- Arrest data show 3.27% are arrested by age 18; birth records show 10.11% of females have a live birth by age 19; labour-market records show 75.52% are formally employed in adulthood.
- Above-median assigned-school students have stronger academic outcomes and lower arrest/teen-motherhood rates than below-median assigned-school students.

Read. Figure 1.

- The top panels show school-specific admission cutoffs within choice groups.
- The bottom-left panel illustrates discontinuity variation around a cutoff.
- The bottom-right panel illustrates across-cutoff variation after adjusting for choice groups.
- Together, these two sources of variation identify relative school effects across many schools.

Economic message. The data are broad enough to study multiple outcomes, and the assignment rule creates quasi-experimental variation in school attendance.

	All schools	Males	Females	Above median	Below median
	(1)	(2)	(3)	(4)	(5)
Panel A: SEA data (cohorts: 1995–2012)					
Female (%)	50.46 (50.00)			53.99 (49.84)	46.94 (49.91)
Admitted cohort size	231.17 (177.65)	233.75 (176.52)	228.78 (178.65)	168.15 (130.07)	325.16 (196.56)
Standardized SEA score	0.00 (1.00)	−0.13 (1.04)	0.13 (0.94)	0.70 (0.64)	−0.70 (0.78)
Individuals	329,481	163,217	166,264	164,519	164,962
Panel B: NCSE data (linked to SEA cohorts: 2006–2012)					
Took NCSE (%)	90.22 (29.70)	88.34 (32.10)	92.10 (26.98)	95.18 (21.42)	82.54 (37.96)
Standardized NCSE score	0.00 (1.00)	−0.22 (1.00)	0.21 (0.96)	0.30 (0.90)	−0.55 (0.94)
Individuals	111,294	55,517	55,777	67,616	43,678
Panel C: CSEC data (linked to SEA cohorts: 1995–2011)					
Took at least 1 subject (%)	73.22 (44.28)	66.96 (47.04)	79.36 (40.47)	86.98 (33.66)	59.78 (49.03)
Number of subjects passed	3.17 (3.09)	2.55 (2.95)	3.78 (3.11)	4.83 (2.96)	1.55 (2.25)
Qualified for tertiary (%) *	34.55 (47.55)	27.53 (44.66)	41.45 (49.26)	58.05 (49.35)	11.61 (32.04)
Individuals	313,580	155,322	158,258	154,921	158,659
Panel D: CAPE data (linked to SEA cohorts: 1999–2009)					
Took at least 1 unit (%)	19.82 (39.86)	15.33 (36.03)	24.24 (42.85)	33.59 (47.23)	4.35 (20.40)
Number of units passed	1.40 (2.93)	1.06 (2.61)	1.73 (3.18)	2.40 (3.54)	0.27 (1.33)
Earned associate degree (%)	14.73 (35.44)	10.87 (31.12)	18.54 (38.87)	25.78 (43.74)	2.32 (15.05)
Earned scholarship (%)	0.95 (9.68)	0.66 (8.11)	1.23 (11.01)	1.78 (13.23)	0.01 (0.84)
Individuals	208,794	103,682	105,112	110,444	98,350
Panel E: Criminal records (linked to SEA cohorts: 1995–2010)—in percent					
Arrested by 18	5.78 (17.79)	5.78 (23.34)	0.81 (8.97)	1.75 (13.10)	4.72 (21.22)
Individuals	297,948	147,544	150,404	145,288	152,660
Panel F: Birth records (linked to SEA cohorts: 2004–2010)—in percent					
Live birth by 19					
Individuals		10.11 (30.15)	6.86 (25.27)	15.19 (35.90)	
Panel G: Labour market data (linked to SEA cohorts: 1995–2006)—in percent					
Formally employed	75.52 (42.99)	79.38 (40.45)	71.77 (45.01)	78.28 (41.24)	73.64 (44.06)
Individuals	160,912	79,319	81,593	65,420	95,492

Summary statistics.

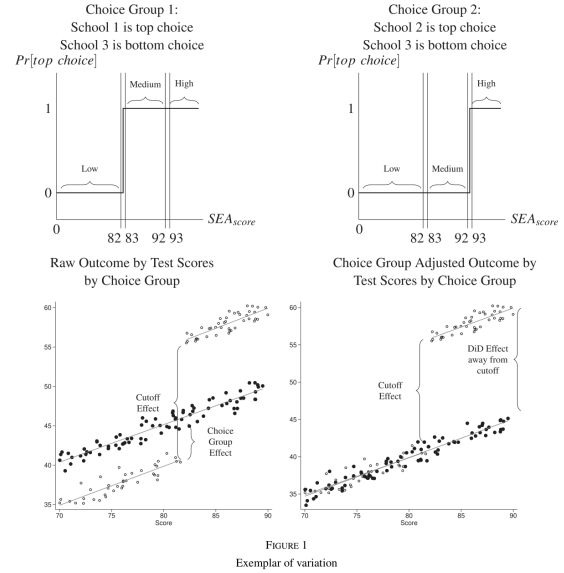


FIGURE 1
Exemplar of variation

Notes: In the top panels, the Y-axis represents the probability of student i being assigned to her top choice; while the X-axis represents the student's SEA score. The top left panel shows the cutoff for School 1 and the top right panel shows the cutoff for School 2. In both panels, those who score below the cutoff for the preferred school are assigned to School 3. This figure illustrates the two different sources of variation. The RD variation identifies the effect of being assigned to School 1 (with respect to School 3) by comparing the outcomes of the Low scoring group (hollow circles right below the cutoff for School 1 in the bottom left panel) to the Medium scoring group (hollow circles right above the cutoff for school 1 in the bottom left panel). This is labelled as "Cutoff Effect." The Difference in Difference variation comes from making comparisons across cutoffs even among students who are away from the cutoff. For example, the difference in outcomes between the Low scoring group at Choice Group 1 (hollow circles in bottom left panel) and the Low scoring group at Choice Group 2 (solid circles in bottom left panel) will reflect differences in choices (as both groups were assigned to the same School 3)—labelled as "Choice Group Effect." Therefore, provided that the effects of choices and test scores on outcomes are additively separable, the Cutoff Effect of attending School 1 vs. 3 can also be identified exploiting observations away from the cutoff after Choice Group Effects are accounted for (graphically shown in the bottom right panel).

Exemplar of variation.

6.2 Table 2: magnitude and multidimensionality of school impacts

Read.

- The standard deviation of persistent school impacts is 0.441 for the high-stakes index and 0.473 for the low-stakes index.
- For binary outcomes, persistent school effects are also meaningful: 0.090 for no dropout by age 14, 0.173 for no live birth by age 19, 0.037 for not arrested by age 18, and 0.070 for formal employment at age 27+.
- The average correlation of high-stakes impacts with low-stakes impacts is only 0.100.
- The average correlation of high-stakes impacts with no live birth by age 19 is negative, while correlations with arrests and employment are positive but modest.
- Even when weighting around the 75th percentile of achievement, high-stakes impacts do not become a universal proxy for all outcomes.

Economic message. A one-dimensional measure of school quality is misleading. Schools can be good at producing test scores and not especially good at producing other outcomes, and vice versa.

6.3 Table 3: inference tests for parent preferences

Read.

TABLE 2
Standard deviations and correlations of persistent school impacts

Outcome	School level (σ_{error})	School level correlations with high-stakes	
	Size of impact	Average	75th %ile of the achievement distribution
	(1)	(2)	(3)
Standardized outcomes			
High-stakes index	0.441 [0.397, 0.483]	1.000	1.000
Low-stakes index	0.473 [0.433, 0.508]	0.100 [-0.042, 0.224]	0.032 [-0.106, 0.177]
Binary outcomes			
No dropout by 14	0.090 [0.077, 0.100]	0.121 [0.019, 0.228]	0.117 [-0.044, 0.290]
No live birth by 19	0.173 [0.147, 0.193]	-0.036 [-0.171, 0.087]	0.174 [0.021, 0.311]
Not arrested by 18	0.037 [0.032, 0.041]	0.282 [0.164, 0.419]	0.407 [0.210, 0.619]
Formally employed 27+	0.070 [0.058, 0.079]	0.152 [0.025, 0.294]	0.014 [-0.148, 0.157]

Notes: All estimates shown were computed by bootstrap with 1,000 repetitions of the maximum likelihood approach described in the text. We report the median as the point estimate, as well as the 5th and 95th percentiles for the confidence intervals. Column (1) reports estimated standard deviations of the persistent school impacts for each outcome. Column (2) reports estimated correlations of the persistent school impacts on the high-stakes index (for the average student) with the persistent school impacts on other outcomes (also for the average student). Column (3) reports estimated correlations of the persistent school impacts on the high-stakes index (estimated with weights centered around the 75th percentile of the achievement distribution) with the persistent school impacts on other outcomes (estimated with weights centered around the 75th percentile of the achievement distribution). We do this using $\text{weight}_i = (1 + \frac{(75 - p_{\text{ct}_i})^2}{100})^{-1}$, where p_{ct_i} is the student's percentile in the incoming achievement distribution. We removed schools with outlier estimated impacts

TABLE 3
Inference tests

	Impacts Only Model			Full Model			
	Pooled average	Above median score	Below median score	Pooled average	Above median score	Below median score	Female Male
	(1)	(2)	(3)	(4)	(5)	(6)	(7) (8)
Standardized value-added on							
High-stakes index	0.620 (0.013)	0.984 (0.022)	0.255 (0.014)	0.217 (0.030)	0.440 (0.048)	-0.005 (0.035)	0.331 (0.047)
Low-stakes index	0.031 (0.009)	0.031 (0.013)	0.029 (0.011)	0.039 (0.009)	0.042 (0.034)	0.035 (0.016)	0.177 (0.033)
No dropout by age 14	0.084 (0.019)	0.119 (0.032)	-0.001 (0.020)	-0.007 (0.017)	0.045 (0.029)	-0.008 (0.016)	0.030 (0.020)
No live birth by age 19	0.081 (0.018)	0.064 (0.010)	0.019 (0.020)	0.116 (0.018)	0.044 (0.032)	0.189 (0.018)	0.045 (0.024)
Not arrested by age 18	0.300 (0.023)	0.532 (0.034)	0.247 (0.031)	0.227 (0.025)	0.337 (0.039)	0.118 (0.031)	0.276 (0.032)
Formally employed at age 27+	0.180 (0.013)	0.264 (0.022)	0.086 (0.014)	0.154 (0.015)	0.240 (0.036)	0.068 (0.014)	0.157 (0.024)

Notes: This table presents pooled averages of the choice model estimates on the standardized value-added of schools (i.e. β_i from equation 12) for two main models: (a) an Impacts Only Model, which includes schools' causal impact estimates for all outcomes, peer quality, and log distance (presenting the aggregated pooled average (Column 1) and the average by incoming achievement level (Columns 2 and 3)) and (b) a Full Model that includes schools' causal impact estimates for all outcomes, the school-level averages for all outcomes, peer quality, and log distance (presenting the aggregated overall average (Column 4) and the averages by incoming achievement level (Columns 5 and 6), and by gender (Columns 7 and 8)). All specifications include control variables for whether the secondary school is on the same island, whether it is all-girls, whether it is all-boys, the estimated likelihoods of school rejection only when comparing the top choice with all unranked schools, and interactions of these likelihoods with all school attributes. The pooled means are computed by taking the average across the point estimates of all cells in a subgroup. Assuming independence across cells, the pooled standard error is computed by summing all subgroup variances, dividing by the number of cells in the subgroup, and taking the squared-root of the ratio. The pooled standard errors are presented in parentheses below each pooled average estimate.

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- In the Impacts Only Model, the pooled coefficient on high-stakes value added is 0.620, with larger responsiveness among above-median-score students.
- In the Full Model, which also includes school averages, peer quality, and other attributes, the pooled high-stakes coefficient remains positive at 0.217.
- Parents also respond to non-academic value added. The Full Model coefficient is 0.227 for not arrested by age 18 and 0.154 for formal employment at age 27+.
- The Full Model coefficient on no live birth by age 19 is 0.116 pooled, but the relationship is stronger below the median achievement score.
- Gender patterns differ: female students' parents are more responsive to low-stakes value added, while both male and female parents respond to arrest and employment impacts.

Economic message. Parents appear to value more than peer quality and average school outcomes. Their rankings are related to causal school impacts on both academic and non-academic dimensions.

6.4 Figures 2 and 3: preferences for academic and non-academic impacts

Read. Figure 2.

- High-achieving students' choices are strongly related to schools' causal impacts on the high-stakes index.
- This relationship remains visible in the Full Model after conditioning on school averages and peer quality.
- Low-stakes test impacts show weaker and less stable patterns.
- Dropout impacts are less consistently valued once the model includes additional school characteristics.

Read. Figure 3.

- Parents are more likely to choose schools that reduce teen arrests.

- Parents are also more likely to choose schools that increase formal employment.
- For no live birth by age 19, the pattern is strongest among lower-achieving students in the Full Model.
- These non-academic relationships are not merely due to test-score value added, because the model includes impacts for all outcomes simultaneously.

Economic message. Families may be responding to broad school quality, not just test-score quality. This helps explain why parent choices sometimes appear weakly related to test-score value added in other settings.

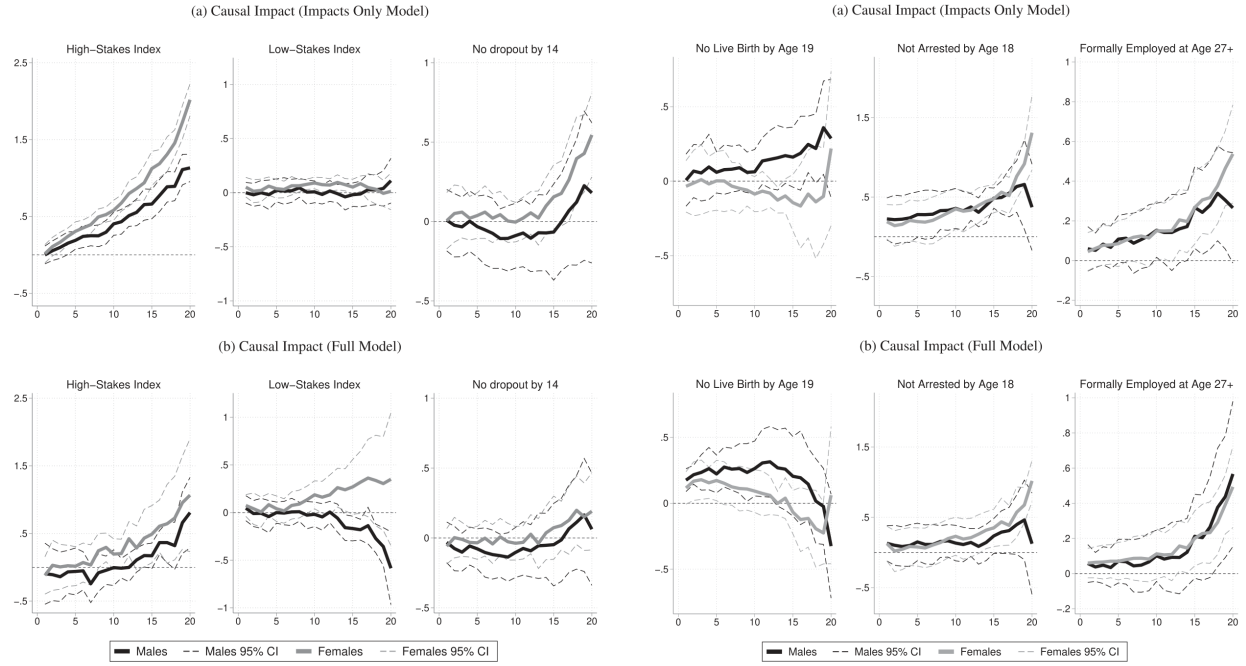


FIGURE 2
Academic outcomes
Notes: The X-axis represent the individual SEA score ventile. The connected lines represent the estimated coefficients, computed separately for each (SEA score ventile) $\times$ (gender) cell, for two main models: (a) displays estimates from the Impacts Only Model, which includes schools' causal impact estimates for all outcomes, peer quality, and log distance and (b) displays estimates from the Full Model, which includes schools' causal impact estimates for all outcomes, the school-level averages for all outcomes, peer quality, and log distance. All specifications include control variables for whether the secondary school is on the coast, whether it is all-boys, whether it is all-girls, whether it is all-boys, whether it is all-girls.

FIGURE 3
Non-academic outcomes
Notes: The X-axis represent the individual SEA score ventile. The connected lines represent the estimated coefficients, computed separately for each (SEA score ventile) $\times$ (gender) cell, for two main models: (a) displays estimates from the Impacts Only Model, which includes schools' causal impact estimates for all outcomes, peer quality, and log distance and (b) displays estimates from the Full Model, which includes schools' causal impact estimates for all outcomes, the school-level averages for all outcomes, peer quality, and log distance. All specifications include control variables for whether the secondary school is on the coast, whether it is all-boys, whether it is all-girls, whether it is all-boys, whether it is all-girls.

Academic outcomes.

Non-academic outcomes.

6.5 Figure 4: relative valuation of non-academic versus test-score impacts

Read.

- Figure 4 compares the strength of preference for non-academic value added relative to high-stakes value added.
- Low-achieving students' parents place relatively more weight on schools that reduce teen motherhood, arrests, and improve employment.
- High-achieving students' parents place relatively more weight on high-stakes test-score impacts.
- The distance-scaled interpretation asks how much additional travel distance parents would accept for a one-standard-deviation improvement in each type of value added.

Economic message. Parent preferences are heterogeneous. School choice may create different competitive pressures across student markets: more test-score pressure in high-achievement markets and more non-academic pressure in low-achievement markets.

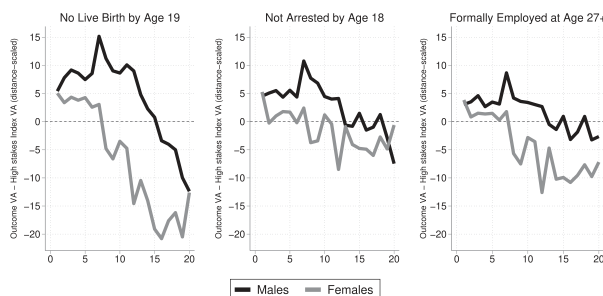


FIGURE 4
Relative comparisons of choice model's estimated coefficients

Notes: This figure presents the difference between the choice model's estimates on the school impacts of three non-academic outcomes and the choice model's estimate on the school impacts of the high-stakes index, scaled by the log distance estimate. The X-axis represents the individual score ventile. The connected lines represent the difference between the choice model estimate on the non-academic impacts and the choice model estimate on the high-stakes impacts divided by the log distance cell estimate (and scaled by 6 for ease of interpretation). This difference is computed separately for each (SEA score ventile) $\times$ (gender) cell. The estimates result from the Full Model, which includes schools' causal impact estimates for all outcomes, the school-level averages for all outcomes, peer quality and log distance, control variables *excluding* the secondary school to which the same student is assigned to in all state schools to which it is off-track, the estimated thresholds of school

Relative comparisons of choice-model coefficients.

7 Section-by-section notes

7.1 Section 1: Introduction

The introduction frames the paper around two questions. First, are test-score impacts a sufficient measure of school quality? Second, do parents choose schools that causally improve outcomes? The authors argue that both questions matter for school choice policy and accountability policy. If school output is multidimensional, then test-based accountability may reward the wrong schools or create incentives to ignore important outcomes. If parents cannot identify causal school impacts, then school choice may not improve productivity in education.

Interpretation. The introduction positions the paper as a bridge between two literatures: school value-added and school choice. The novelty is that both are studied together using a common quasi-experimental design.

7.2 Section 2: Context and data

The paper describes Trinidad and Tobago as a centralized school-choice environment where nearly all students apply to public secondary schools through the same system. Families submit ranked lists, and assignments depend on choices, SEA scores, school capacities, and institutional rules. The authors then describe the administrative linkages that allow them to observe outcomes many years later.

Interpretation. The data are powerful because they combine preferences, assignment, attendance, and outcomes. Many school-choice studies observe either preferences or outcomes, but not both at the national administrative scale.

7.3 Section 3: Estimating school impacts

Section 3 is the identification core. The authors define school value added, describe the variation from the assignment algorithm, and estimate school-specific treatment-on-the-treated effects using assignment as an instrument for attendance. The model includes choice-group fixed effects and flexible controls for SEA scores. The paper then validates the identifying assumptions through balance tests, consistency between RD and across-cutoff variation, and additivity tests.

Interpretation. The strength of the empirical design is that it uses the assignment mechanism itself. The paper is not simply comparing students who attend different schools; it is comparing students whose assignment differs because of algorithmic cutoffs and rankings.

7.4 Section 4: Magnitude and multidimensionality of school impacts

Section 4 asks whether school effects are large and whether they line up across outcomes. The paper first estimates the dispersion of persistent school effects using even-odd cohort covariance. It then studies correlations among school impacts. The key result is that schools have large impacts across many outcomes, but impacts are only modestly correlated across dimensions.

Interpretation. This section is the empirical foundation for the multidimensionality claim. If all school effects were highly correlated, test-score value added would be an adequate measure. The results show that it is not.

7.5 Section 5: Estimating preferences for schools

Section 5 links school impacts to parent choices. Because parents submit limited rank-ordered lists under Deferred Acceptance, standard rank-order assumptions may fail. The paper modifies the exploded multinomial logit model to allow for admission probabilities and strategic choice. The estimates show that parents choose schools with causal impacts on high-stakes tests and also schools with causal impacts on arrests, teen motherhood, and employment.

Interpretation. The preference results do not require parents to observe value-added estimates. Parents may infer school quality through reputation, local information, peer networks, or observed average outcomes. The key point is that their choices are systematically related to causal school impacts.

7.6 Section 6: Conclusions and policy implications

The conclusion emphasizes that school quality is multidimensional and that parents may value multiple dimensions. It warns that test-score accountability can miss important outcomes and that school choice may work differently for different student groups. The paper also suggests that providing information on causal impacts across academic and non-academic outcomes could improve parental choices.

Interpretation. The policy lesson is not to abandon testing or school choice. It is to broaden the measurement of school quality and to understand which dimensions families value.

8 Short summary

- The paper studies whether school value added is multidimensional and whether parents choose schools based on causal impacts.
- The setting is Trinidad and Tobago’s centralized secondary-school assignment system.
- The data link school choices and assignments to academic exams, arrests, teen births, and formal employment.
- The identification strategy uses algorithm-based assignment as an instrument for actual school attendance.
- The paper estimates causal effects for 132 public secondary schools.
- Schools have meaningful impacts on tests, dropout, teen motherhood, arrests, and employment.
- High-stakes test value added is only weakly correlated with many non-test outcomes.
- Parents choose schools that improve high-stakes tests, even after controlling for peer quality and average outcomes.
- Parents also choose schools that reduce arrests and teen motherhood and increase formal employment.
- Low-achieving students’ parents put relatively more weight on non-academic impacts; high-achieving students’ parents put relatively more weight on test-score impacts.
- The incremental contribution is the combination of representative causal school effects, multidimensional outcomes, parent preferences, and a strategic-choice model under limited Deferred Acceptance.

9 Conclusion

9.1 Core conclusion

Beuermann, Jackson, Navarro-Sola, and Pardo show that a good school cannot be defined by test-score value added alone. Individual schools causally affect multiple outcomes, and the schools that improve high-stakes tests are not always the same schools that reduce crime, reduce teen motherhood, or improve formal employment.

9.2 Economic intuition

Schools produce multiple kinds of human capital. Some schools may be especially good at academic preparation, while others may be especially good at discipline, social behaviour, engagement, or labour-market readiness. If families have information about these dimensions, their school choices can reflect more than test scores.

9.3 Policy takeaway

The paper suggests two reforms. First, accountability systems should include broader measures of school value added, not only high-stakes exams. Second, school-choice systems could be improved by giving families information about schools' causal impacts across multiple outcomes. The central message is that test-score-based school quality is incomplete, and parent preferences may be more multidimensional than standard models assume.